Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Multiplying and Dividing Terms

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KEY WORDS term *(tūnga)* · product *(otinga whakarea)* · power *(taupū)* · simplify *(whakamāmā)*

A Multiplying terms ●○○

Remember: Multiply the numbers, then write the letters next to each other in alphabetical order: $3a \times 2b = 6ab$.

1. Simplify.

a) $3 \times 2a$ = _____ **b)** $4a \times 5$ = _____ **c)** $2a \times 3b$ = _____

d) $5m \times 2n$ = _____ **e)** $x \times y$ = _____ **f)** $-2a \times 6b$ = _____

g) $7p \times (-3q)$ = _____ **h)** $-4x \times (-5y)$ = _____ **i)** $2a \times 3b \times 4c$ = _____

j) $0.5x \times 8y$ = _____ **k)** $6 \times (-2k)$ = _____ **l)** $3ab \times 5c$ = _____

m) $9g \times 4$ = _____ **n)** $-a \times 7b$ = _____ **o)** $10x \times 10y$ = _____

p) $2p \times (-2q) \times (-2r)$ = _____ **q)** $1.5m \times 4n$ = _____ **r)** $-8 \times (-3d)$ = _____

s) $5xy \times 3z$ = _____ **t)** $4h \times 0.25k$ = _____

B Multiplying with powers ●●○

Remember: A letter multiplied by itself is a power: $a \times a = a^2$. When you multiply powers of the same letter, add the powers: $a^2 \times a^3 = a^5$.

2. Simplify.

a) $a \times a$ = _____ **b)** $3x \times 4x$ = _____ **c)** $a^2 \times a^3$ = _____ **d)** $2x^2 \times 5x$ = _____

e) $-3y \times 2y$ = _____ **f)** $4m^3 \times 2m^2$ = _____ **g)** $3ab \times 2a$ = _____ **h)** $5x^2y \times 2xy$ = _____

i) $(3x)^2$ = _____ **j)** $(2a)^3$ = _____ **k)** $(-4y)^2$ = _____ **l)** $2a \times 3a \times a$ = _____

m) $x^3 \times x$ = _____ **n)** $6n \times (-2n^2)$ = _____ **o)** $2pq \times 3pq$ = _____ **p)** $(10k)^2$ = _____

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C Dividing terms ●●○

Remember: Divide the numbers, then cancel any letters that are on both the top and the bottom: $\frac{12ab}{4a} = 3b$.

3. Simplify.

a) $\frac{12a}{4}$ **b)** $\frac{15xy}{3x}$ **c)** $\frac{8a^2}{2a}$ **d)** $\frac{20x^3}{5x^2}$

e) $\frac{18ab}{6b}$ **f)** $\frac{-24m}{8}$ **g)** $\frac{9x^2y}{3xy}$ **h)** $\frac{30a^3b}{10a}$

i) $\frac{14pq}{-2p}$ **j)** $\frac{16x^4}{4x}$ **k)** $\frac{6ab}{6ab}$ **l)** $\frac{-36x^2}{-9x}$

m) $\frac{45k}{9k}$ **n)** $\frac{21m^2n}{7m}$ **o)** $\frac{50x^5}{10x^3}$ **p)** $\frac{-16abc}{4ac}$

q) $\frac{27y^3}{3y^3}$ **r)** $\frac{100pq^2}{25q}$ **s)** $\frac{8x^2y^2}{8xy}$ **t)** $\frac{-60a^4}{12a^2}$

D Mixed practice ●●●

4. Simplify.

a) $\frac{6x \times 2y}{4x}$ **b)** $\frac{3a \times 8b}{6}$ **c)** $\frac{5x^2 \times 4x}{10x}$

d) $2a^2b \times 3ab^2$ **e)** $\frac{28x^3y^2}{7xy}$ **f)** $(2ab)^2$

g) $\frac{9m \times (-2n)}{3m}$ **h)** $\frac{4p^2 \times 3q}{6p}$ **i)** $-x \times (-x) \times (-x)$

E Finding the missing term ●●●

Remember: Divide the answer by the term you know.

5. Fill in the box.

a) $4a \times \square = 12ab$ **b)** $3x \times \square = 15x^2$ **c)** $2 \times \square = 10pq$

d) $5m \times \square = -20mn$ **e)** $6x \times \square = 18x^3$ **f)** $a \times \square = 7a^2b$

g) $-2y \times \square = 8y^2$ **h)** $9k \times \square = 9k$

★ Extension

6. A rectangle is $3x$ long and $2y$ wide. Write an expression for its area.

7. A cube has edges of length $2a$. Write an expression for its volume.

8. Find the missing term: $4a \times \square = 12ab$.

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ALGEBRA

Multiplying and Dividing Terms

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KEY WORDS term *(tūnga)* · product *(otinga whakarea)* · power *(taupū)* · simplify *(whakamāmā)*

A Multiplying terms ●○○

Remember: Multiply the numbers, then write the letters next to each other in alphabetical order: $3a \times 2b = 6ab$.

1. Simplify.

a) $4 \times 3a$ = _____ **b)** $2a \times 7$ = _____ **c)** $3a \times 4b$ = _____

d) $6m \times 2n$ = _____ **e)** $p \times q$ = _____ **f)** $-3a \times 5b$ = _____

g) $8p \times (-2q)$ = _____ **h)** $-6x \times (-3y)$ = _____ **i)** $3a \times 2b \times 5c$ = _____

j) $0.5x \times 6y$ = _____ **k)** $7 \times (-3k)$ = _____ **l)** $4ab \times 6c$ = _____

m) $8g \times 5$ = _____ **n)** $-a \times 9b$ = _____ **o)** $10p \times 10q$ = _____

p) $3p \times (-q) \times (-3r)$ = _____ **q)** $2.5m \times 4n$ = _____ **r)** $-9 \times (-2d)$ = _____

s) $6xy \times 2z$ = _____ **t)** $8h \times 0.25k$ = _____

B Multiplying with powers ●●○

Remember: A letter multiplied by itself is a power: $a \times a = a^2$. When you multiply powers of the same letter, add the powers: $a^2 \times a^3 = a^5$.

2. Simplify.

a) $b \times b$ = _____ **b)** $2x \times 6x$ = _____ **c)** $a^3 \times a^4$ = _____ **d)** $3x^2 \times 4x$ = _____

e) $-2y \times 5y$ = _____ **f)** $5m^2 \times 2m^4$ = _____ **g)** $4ab \times 2b$ = _____ **h)** $3x^2y \times 4xy$ = _____

i) $(5x)^2$ = _____ **j)** $(3a)^3$ = _____ **k)** $(-2y)^2$ = _____ **l)** $3a \times 2a \times a$ = _____

m) $x^4 \times x$ = _____ **n)** $5n \times (-3n^2)$ = _____ **o)** $4pq \times 2pq$ = _____ **p)** $(6k)^2$ = _____

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C Dividing terms ●●○

Remember: Divide the numbers, then cancel any letters that are on both the top and the bottom: $\frac{12ab}{4a} = 3b$.

3. Simplify.

a) $\frac{15a}{5}$ **b)** $\frac{12xy}{4x}$ **c)** $\frac{10a^2}{2a}$ **d)** $\frac{24x^3}{6x^2}$

e) $\frac{20ab}{4b}$ **f)** $\frac{-28m}{7}$ **g)** $\frac{12x^2y}{4xy}$ **h)** $\frac{40a^3b}{10a}$

i) $\frac{18pq}{-3p}$ **j)** $\frac{20x^4}{5x}$ **k)** $\frac{9ab}{9ab}$ **l)** $\frac{-32x^2}{-8x}$

m) $\frac{42k}{7k}$ **n)** $\frac{27m^2n}{9m}$ **o)** $\frac{60x^5}{10x^3}$ **p)** $\frac{-18abc}{6ac}$

q) $\frac{24y^3}{4y^3}$ **r)** $\frac{80pq^2}{20q}$ **s)** $\frac{5x^2y^2}{5xy}$ **t)** $\frac{-48a^4}{12a^2}$

D Mixed practice ●●●

4. Simplify.

a) $\frac{8x \times 3y}{6x}$ **b)** $\frac{4a \times 9b}{6}$ **c)** $\frac{6x^2 \times 5x}{10x}$

d) $3a^2b \times 2ab^2$ **e)** $\frac{36x^3y^2}{9xy}$ **f)** $(3ab)^2$

g) $\frac{8m \times (-3n)}{4m}$ **h)** $\frac{6p^2 \times 2q}{4p}$ **i)** $-y \times (-y) \times (-y)$

E Finding the missing term ●●●

Remember: Divide the answer by the term you know.

5. Fill in the box.

a) $5a \times \square = 15ab$ **b)** $4x \times \square = 12x^2$ **c)** $3 \times \square = 12pq$

d) $2m \times \square = -14mn$ **e)** $5x \times \square = 20x^3$ **f)** $b \times \square = 6ab^2$

g) $-3y \times \square = 9y^2$ **h)** $7k \times \square = 7k$

★ Extension

6. A rectangle is $4x$ long and $3y$ wide. Write an expression for its area.

7. A cube has edges of length $3a$. Write an expression for its volume.

8. Find the missing term: $5x \times \square = 15xy$.

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